CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (IT) Examination

May 2013

IT404 Language Processor

Date: 11.05.2013, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1	**(a)**	What is language processor? Explain types of language processor.	**[04]**
	(b)	Differentiate between 1) Compiler and Interpreter 2) Token and Lexeme	**[03]**
Q - 2	**(a)**	List out the different phases of compiler. Explain significance of each phase of compiler with an example.	**[08]**
	(b)	Draw DFA for following RE (111+ 100)*0	**[04]**
	(c)	Explain synthesized attribute and inherited attribute.	**[02]**
		OR	
Q - 2	**(a)**	Draw NFA- ^ for following regular expression (a/b)*abb Construct DFA from NFA. Also prepare transition table.	**[08]**
	(b)	Explain two pass assembly in detail.	**[04]**
	(c)	List out the different types of errors in compiler and give an example of each.	**[02]**
Q - 3	**(a)**	Explain the role of the lexical analyzer.	**[06]**
	(b)	What is symbol table? Why it is used? Which phase of compiler set the attributes of token in a symbol table?	**[05]**
	(c)	Explain the single pass and multi pass compiler.	**[03]**
		OR	
Q - 3	**(a)**	Explain Error Recovery Strategies in detail with suitable example.	**[06]**
	(b)	What is activation record? Explain clearly the components of an activation record.	**[05]**
	(c)	Check whether the below grammar is Ambiguous or not? Which language is generated from this grammar? $S \rightarrow aSbS \mid bSa \mid \varepsilon$	**[03]**

SECTION – II

Q - 4 (a) Define following terms: **[08]**

1) Left recursion
2) Handle pruning
3) Three address code
4) S-R conflicts

Q - 5 (a) Generate Canonical LR(0) item set for following grammar **[05]**

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L$

(b) Compute FIRST and FOLLOW set for following **[04]**

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

$F \rightarrow (E) \mid id$

(c) Write Quadruples for **[04]**

(a+b)*(c+d)-(a+b+c)

OR

Q - 5 (a) Check following grammar is LL(1) or not. **[05]**

$S' \rightarrow S\#$

$S \rightarrow aSA \mid \varepsilon$

$A \rightarrow bS \mid \varepsilon$

(b) What is operator grammar? Write advantages and disadvantages of operator precedence parsing. **[04]**

(c) Perform the Left factoring of following Grammar **[04]**

$A \rightarrow ad \mid a \mid ab \mid abc \mid b$

Q - 6 (a) List out the different code optimization techniques. **[02]**

(b) **Attempt any Three.** **[12]**

1) Explain the functions of linker and loader.
2) Compare static memory allocation with dynamic memory allocation.
3) Preprocessor
4) Compare Top Down and Bottom Up parser.

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